JavaScript is disabled on your browser.

compact1, compact2, compact3

java.lang.annotation

Enum ElementType

java.lang.Object
    java.lang.Enum<ElementType>
        java.lang.annotation.ElementType

All Implemented Interfaces:

Serializable, Comparable<ElementType>

```
public enum ElementType
extends Enum<ElementType>
```

The constants of this enumerated type provide a simple classification of the syntactic locations where annotations may appear in a Java program. These constants are used in `java.lang.annotation.Target` meta-annotations to specify where it is legal to write annotations of a given type.

The syntactic locations where annotations may appear are split into *declaration contexts* , where annotations apply to declarations, and *type contexts* , where annotations apply to types used in declarations and expressions.

The constants `ANNOTATION_TYPE` , `CONSTRUCTOR` , `FIELD` , `LOCAL_VARIABLE` , `METHOD` , `PACKAGE` , `PARAMETER` , `TYPE` , and `TYPE_PARAMETER` correspond to the declaration contexts in JLS 9.6.4.1.

For example, an annotation whose type is meta-annotated with `@Target(ElementType.FIELD)` may only be written as a modifier for a field declaration.

The constant `TYPE_USE` corresponds to the 15 type contexts in JLS 4.11, as well as to two declaration contexts: type declarations (including annotation type declarations) and type parameter declarations.

For example, an annotation whose type is meta-annotated with `@Target(ElementType.TYPE_USE)` may be written on the type of a field (or within the type of the field, if it is a nested, parameterized, or array type), and may also appear as a modifier for, say, a class declaration.

The TYPE_USE constant includes type declarations and type parameter declarations as a convenience for designers of type checkers which give semantics to annotation types. For example, if the annotation type NonNull is meta-annotated with @Target(ElementType.TYPE_USE), then @NonNull class C {...} could be treated by a type checker as indicating that all variables of class C are non-null, while still allowing variables of other classes to be non-null or not non-null based on whether @NonNull appears at the variable's declaration.

Since:

1.5

See *The Java™ Language Specification*:

9.6.4.1 @Target, 4.1 The Kinds of Types and Values

Enum Constant Summary

Enum Constants

Enum Constant	Description
ANNOTATION_TYPE	Annotation type declaration
CONSTRUCTOR	Constructor declaration
FIELD	Field declaration (includes enum constants)
LOCAL_VARIABLE	Local variable declaration
METHOD	Method declaration
PACKAGE	Package declaration
PARAMETER	Formal parameter declaration
TYPE	Class, interface (including annotation type), or enum declaration
TYPE_PARAMETER	Type parameter declaration
TYPE_USE	Use of a type

Method Summary

Concrete Methods

Modifier and Type	Method	Description
static ElementType	valueOf(String name)	

		Returns the enum constant of this type with the specified name.
static ElementType[]	values()	Returns an array containing the constants of this enum type, in the order they are declared.

Methods inherited from class java.lang.Enum

clone, compareTo, equals, finalize, getDeclaringClass, hashCode, name, ordinal, toString, valueOf

Methods inherited from class java.lang.Object

getClass, notify, notifyAll, wait, wait, wait

Enum Constant Detail

TYPE

```
public static final ElementType TYPE
```

Class, interface (including annotation type), or enum declaration

FIELD

```
public static final ElementType FIELD
```

Field declaration (includes enum constants)

METHOD

```
public static final ElementType METHOD
```

Method declaration

PARAMETER

```
public static final ElementType PARAMETER
```

Formal parameter declaration

CONSTRUCTOR

```
public static final ElementType CONSTRUCTOR
```

Constructor declaration

LOCAL_VARIABLE

```
public static final ElementType LOCAL_VARIABLE
```

Local variable declaration

ANNOTATION_TYPE

```
public static final ElementType ANNOTATION_TYPE
```

Annotation type declaration

PACKAGE

```
public static final ElementType PACKAGE
```

Package declaration

TYPE_PARAMETER

```
public static final ElementType TYPE_PARAMETER
```

Type parameter declaration

Since:

1.8

TYPE_USE

```
public static final ElementType TYPE_USE
```

Use of a type

Since:

1.8

Method Detail

values

```
public static ElementType[] values()
```

Returns an array containing the constants of this enum type, in the order they are declared. This method may be used to iterate over the constants as follows:

```
for (ElementType c : ElementType.values())
    System.out.println(c);
```

Returns:

an array containing the constants of this enum type, in the order they are declared

valueOf

```
public static ElementType valueOf(String name)
```

Returns the enum constant of this type with the specified name. The string must match *exactly* an identifier used to declare an enum constant in this type. (Extraneous whitespace characters are not permitted.)

Parameters:

name - the name of the enum constant to be returned.

Returns:

the enum constant with the specified name

Throws:

IllegalArgumentException - if this enum type has no constant with the specified name

NullPointerException - if the argument is null

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms.
Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.